



CD/JUL 2024/ICT450/ITS432

**UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION**

COURSE	:	DATABASE DESIGN AND DEVELOPMENT
COURSE CODE	:	ICT450/ITS432
EXAMINATION	:	JULY 2024
TIME	:	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of five (5) questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
 - i. the Question Paper
 - ii. an Answer Booklet – provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages

QUESTION 1

- a) A database management system (DBMS) is a software tool that support various activities and functions for data management. Briefly explain **TWO (2)** functions of DBMS that guarantee integrity and consistency of data.

(4 marks)

- b) Identify **FOUR (4)** examples of database applications usage.

(4 marks)

- c) Table 1 shows an example of data from Gerai 17's database system. The owner of Gerai 17, Cik Siti, is having difficulty getting the total number of menu and average price of her menu.

Table 1: MENU Table

Menu Code	Menu Name	Price (RM)
M1	Nasi Bujang	3.50
M2	Nasi Ayam	
	Bihun Goreng	3.00
M4	Laksa	2.50

Identify **THREE (3)** problems exist in Table 1. Justify your answer.

(6 marks)

- d) Database system is designed based on two design philosophies which are centralized and decentralized. Explain **THREE (3)** differences between these two designs.

(6 marks)

QUESTION 2

- a) The evolution of data models has been driven by the increasing complexity of data storage, management, and analysis needs over time. Match the suitable data models with their correct statements.

Hierarchical	○	○ Visual representation makes it an effective communication tool.
Network	○	○ It may promote islands of information problems as individuals and departments can easily develop their own applications.
Relational	○	○ Parent/child relationship promotes data integrity.
Entity Relationship	○	○ It handles more relationship types, such as M:N and multiparent

(4 marks)

- b) The following relational schema provides a basic structure for managing pet sales in a pet shop system. It includes customer information, pet information and purchase transactions.

PETS (PetID, Name, Species, Age, Price, PurchaseID*)
 CUSTOMERS (CustomerID, Name, Address, Phone, Email)
 PURCHASES (PurchaseID, CustomerID*, PurchaseDate)

Based on the relational schema above, describe **TWO (2)** related business rules.

(4 marks)

- c) Security is a crucial aspect of DBMS. Describe **THREE (3)** reasons for protecting data from unauthorized users.

(6 marks)

- d) Briefly explain **THREE (3)** challenges in handling multimedia data compared to traditional structured data.

(6 marks)

QUESTION 3

Figure 1 shows a Sales Invoice for a new start-up company, Sasa eStore. This invoice is manually generated from Microsoft Excel.

		Sasa eStore	
		Lot 10, Jalan Parameswara 06-2226666 hq@sasa.com	
BILL TO		INVOICE # 100	
Adilla Yusof (Member ID : SS11)		Date: 26/02/2024	
No. 12, Jalan Seringin 8,			
Desa Seringin, 71800 Nilai, Negeri Sembilan.			

Item Name	Quantity	Unit Price	Amount
Green Deep Lip & Eye Remover	1	65	\$65.00
Argan Oil Cleansing Foam	1	49	\$49.00
10% Niacinamide Serum	1	56	\$56.00
Alaska Volcano Clay Mask Stick	2	44	\$88.00
Argan Oil Cleansing Foam	1	49	\$49.00
TOTAL			\$307.00
Paid Amount			\$310.00
Balance Due			\$3.00

Make all checks payable to Sasa eStore

If you have any questions concerning this invoice, use the following contact information:
Azlina, 012-1234567, azlina@sasa.com

THANK YOU FOR YOUR BUSINESS!

Figure 1: Sasa eStore Sales Invoice

Based on the information given, answer the following questions:

- State the normal form of the Figure 1. Explain your answer with reasons. (3 marks)
- Identify the primary key(s) and partial dependencies and/or transitive dependencies. (6 marks)
- Normalize this invoice form from the First Normal Form (1NF) until the Third Normal Form (3NF). List all the steps involved. (11 marks)

QUESTION 4

The relational schema below shows the e-Voting database system for one of the local universities in Malaysia. Each student can vote up to two candidates.

STUDENT (StudentID, FirstName, LastName, PhoneNumber, FacultyID*)

CANDIDATE (CandidateID, StudentID*)

VOTING (VotingID, VotingDate, StudID*)

VOTING_DETAILS (VotingID*, CandidateID*, VotingTime)

FACULTY (FacultyID, FacultyName)

Write the SQL statements based on the following requirements:

- a) Calculate the total number of candidates for each faculty. (4 marks)
- b) Display the details of student who do not vote for any candidate. (4 marks)
- c) Identify the total number of voters for each faculty. (4 marks)
- d) Display the full name of candidates that get more than 500 votes. (4 marks)
- e) Identify the total number of student that vote only one candidate. (4 marks)

QUESTION 5

S2 football club plans to develop a database management system to facilitate the organization and administration of football tournaments at various levels. The system will be used to manage tournament fixtures, teams, players, match results, and venue information. As a database designer, you have been hired by S2 football club to design the database system according to the following requirements:

- Tournaments store details of a tournament ID, tournament name, start date, end date, and organizer information.
- Team refers to football teams who participate in tournaments which include data about team id, team name and coach name.
- Player represents individual football players. Players' details include player id, player name, date of birth, nationality, and position. One player can join only one team.
- Fixture records the whole schedule of match to be played in the tournament. It includes the fixture ID, kick-off time and tournament ID.
- Match contains information about individual matches which include data for match id, match date, home team ID, away team ID, result and venue.
- Venue stores details of selected venues for each matches including venue id, venue name, location, and seating capacity.
- A tournament can have multiple fixtures, but each fixture belongs to only one tournament.
- A team can have multiple players, but each player belongs to only one team.
- Each fixture consists of multiple matches, but each match is associated with only one fixture.

Based on the given information:

- a) Construct an Entity Relationship Diagram (ERD) complete with all the required tables, attributes, relationships as well as the cardinalities using the Crow's Foot notation that reflects all the given business rules.
- (16 marks)
- b) Identify **FOUR (4)** relevant information that you can get from the ERD you constructed in (a).

(4 marks)

END OF QUESTION PAPER